Project Report

1. Project Title

Gear Train & Power Transmission Calculator

2. Project Objective

The purpose of this project was to better understand how mechanical systems transfer motion and power through gear mechanisms.

By combining mathematics, physics, and programming, I wanted to create a simple engineering tool that demonstrates how gear ratios influence rotational speed, torque, and overall system performance.

3. Research Question

How do gear ratios affect rotational speed (RPM), torque, and power transmission in mechanical systems?

4. Background

Gears are among the most widely used mechanical components in engineering. They are found in automobiles, aircraft systems, industrial machinery, and countless other applications.

Although gears may appear simple, they play a critical role in controlling speed and torque. By changing gear ratios, engineers can design systems that prioritize either speed or force depending on the desired application.

This project explores these relationships through mathematical calculations and engineering analysis.

5. Methodology

The calculator was developed using Python.

The user enters:

- Driver Gear Teeth
- Driven Gear Teeth
- Input RPM
- Input Torque

The program then calculates:

- Gear Ratio
- Output RPM
- Output Torque

The calculations are based on standard mechanical engineering relationships.

Gear Ratio

Gear Ratio = Teeth Driven / Teeth Driver

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Output RPM

Output RPM = Input RPM / Gear Ratio

Output Torque

Output Torque = Input Torque × Gear Ratio

6. Results

The calculator successfully analyzes the effect of gear ratios on speed and torque.

Example

Driver Gear Teeth = 12

Driven Gear Teeth = 48

Input RPM = 1200

Input Torque = 10 Nm

Output

Gear Ratio = 4:1

Output RPM = 300

Output Torque = 40 Nm

The simulation demonstrates how increasing gear ratio decreases rotational speed while increasing torque.

7. Skills Developed

- Python Programming
- Mechanical Engineering Fundamentals
- Engineering Calculations
- Mathematical Modeling
- Systems Thinking
- Data Analysis
- Problem Solving
- Critical Thinking

8. What I Learned

One of the most interesting things I learned from this project is that small mechanical changes can have a major impact on system performance.

Before working on this calculator, I understood the basic idea of gears, but I did not fully appreciate how strongly gear ratios affect speed and torque.

Building this project helped me better understand how engineers use mathematics to design machines, improve performance, and solve practical problems.

I also learned how programming can be used to model mechanical systems and visualize engineering concepts.

9. Future Improvements

Possible future developments include:

- Multi-Stage Gear Systems
- Gear Efficiency Calculations
- Power Loss Analysis
- Interactive Mechanical Dashboard
- Mechanical System Visualization
- Real-World Vehicle Transmission Simulation

10. Personal Reflection

Before starting this project, I mostly thought of gears as simple machine components.

As I learned more about gear ratios, RPM, torque, and power transmission, I realized how important gears are in engineering design. What appears to be a simple mechanism can completely change how a machine behaves.

The most rewarding part of this project was seeing how a few mathematical equations could explain the behavior of real mechanical systems.

It reminded me that engineering is not only about building machines, but also about understanding how systems work and how simple ideas can be used to solve complex problems.

Engineering is the art of transforming motion.